CAIMIRA / VISUAL ABSTRACT

Airborne transmission risk

Close contact + shared air, in one model

Integrating short- and long-range inhalation exposure

1



Ventilation • filtration • settling • viral decay

Emission → Exposure over time → Absorbed dose → Infection risk

Total dose = short-range dose + long-range dose

Monte Carlo sampling captures variability and uncertainty



PATIENT WARD

~10×

Higher short-range dose

95 vs 9 IRP: short vs long range



WELL-FITTED FFP2

13× lower dose

Patient ward: 104 → 8 IRP



SHARED OFFICE

8h ~ 15 min

Comparable modelled risk

Poorly ventilated shared air
vs unmasked face-to-face conversation

Assess proximity and time together. Combine controls.

Source control
Reduce emitted virus

Clean indoor air
Reduce background exposure

Respirators
Reduce inhaled dose

Simulated settings: office • classroom • patient ward • ICU | Event risk also depends on infection prevalence and occupancy.

Scenario-specific model estimates for SARS-CoV-2 Omicron; not universal risk ratios. IRP = infectious respiratory particle.

Henriques et al. (2025) - J. R. Soc. Interface 22:20240740 - doi:10.1098/rsif.2024.0740

CAiMIRA · AIRBORNE TRANSMISSION

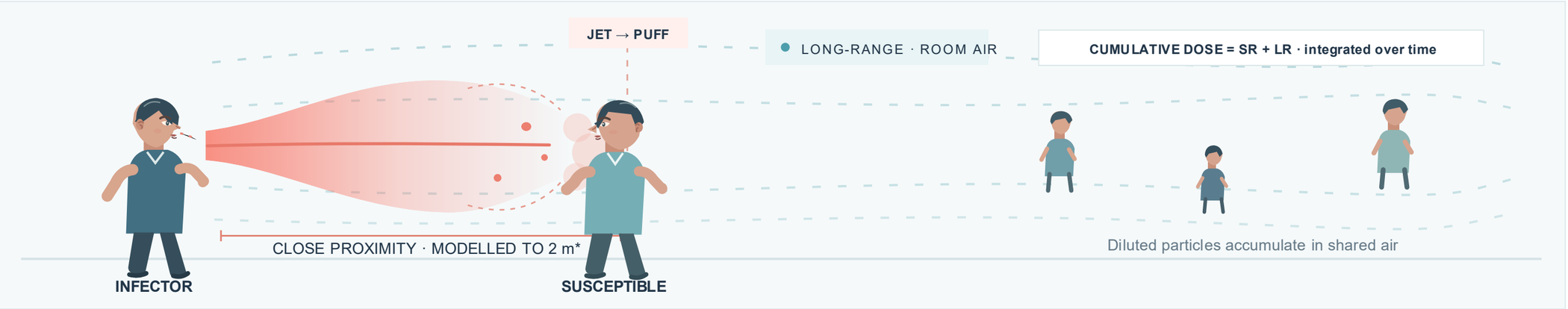
One exposure. Two ranges. One integrated risk model.

A two-stage exhaled jet is coupled to room-air concentration to estimate cumulative inhaled dose.

SHORT + LONG RANGE

THE MULTI-BOX CONTINUUM

Exposure shifts from a concentrated exhaled jet to the shared background air.



MODEL PIPELINE EMISSION — TWO-STAGE JET + ROOM AIR — INHALED DOSE — MONTE CARLO → RISK

* 2 m is an adjustable model boundary

PATIENT WARD · PER NURSE SHIFT

95 **IRP short-range**

vs 9 IRP long-range

Close-proximity dose is about 10× the room-air dose.

RESPIRATOR · SAME WARD SCENARIO

13× lower inhaled dose

with well-fitted FFP2

104 → 8 IRP total dose

SHARED OFFICE · EXPOSURE-TIME MATTERS

8 h in room air

≈ 15 min face-to-face

6 IRP long-range ≈ 8 IRP short-range dose.

Henriques A et al. · J. R. Soc. Interface 22 (2025), 20240740 · DOI: 10.1098/rsif.2024.0740

Model extends CAiMIRA · WHO short- and long-range airborne terminology

CAiMIRA · AIRBORNE TRANSMISSION

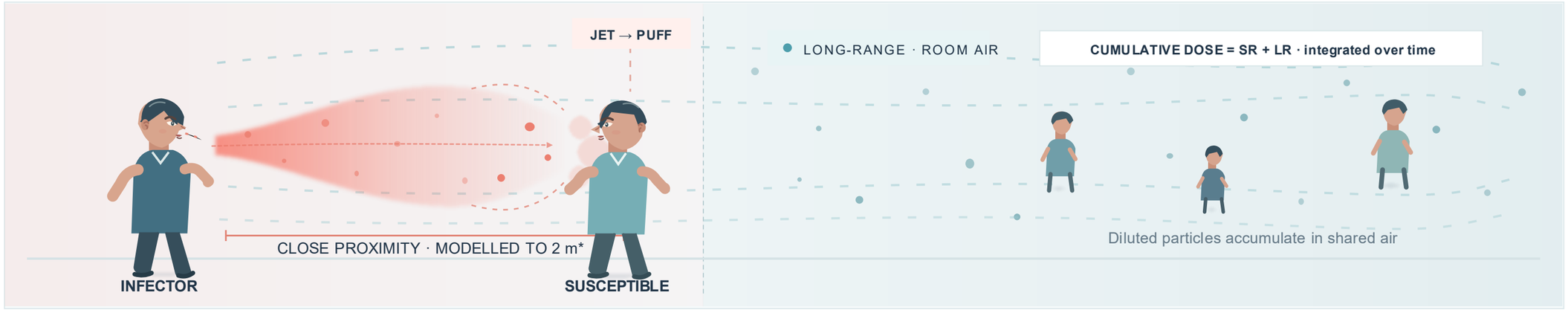
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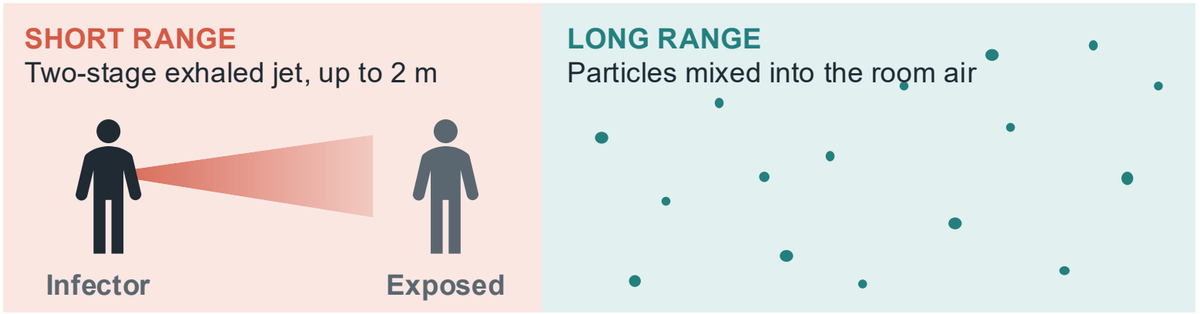
Model extends CAiMIRA · WHO short- and long-range airborne terminology

CAiMIRA · VISUAL ABSTRACT

Close contact and shared room air both drive airborne infection risk

A two-stage exhaled-jet model is coupled to room-air concentration to give one cumulative inhaled dose, tested in four indoor settings

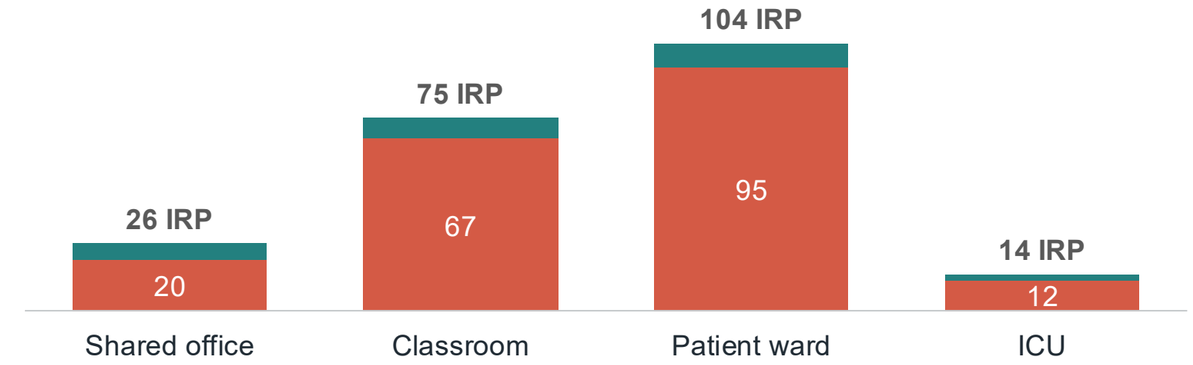
HOW THE MODEL WORKS



Inhaled dose = short-range dose + long-range dose

Monte Carlo sampling of emission, ventilation, distance and masks → infection probability

Mean inhaled dose per exposed person by setting, IRP



PATIENT WARD · ONE NURSE SHIFT

10×

higher dose from close contact than from room air

95 IRP short-range vs 9 IRP long-range

RESPIRATORS · SAME WARD

13×

lower inhaled dose with a well-fitted FFP2

Shift dose falls from 104 IRP to 8 IRP

SHARED OFFICE · EXPOSURE TIME

8 h ≈ 15 min

of poorly ventilated air ≈ one unmasked close talk

6 IRP long-range over 8 h vs 8 IRP short-range in 15 min

Model estimates for SARS-CoV-2 (Omicron), unmasked occupants unless stated; IRP = infectious respiratory particle.

Source: Henriques A et al. 2025, J. R. Soc. Interface 22: 20240740, doi:10.1098/rsif.2024.0740

Compare

	1 - Image-based	2 - SVG-based into shapes	3 - SVG-based into shapes + Claude AI pptx integration with prompt “improve this SVG graphical abstract.”	4 - Claude AI pptx integration with prompt “create a visual abstract slide on this paper”
Quality of the illustration	Very good quality. Neater as it is pixel based and actucally took less generation time then the SVG option	Clean, consistent vector layout, but flatter than the image. People figures look crude and distorted.	Same layout as 2, with key numbers in colour and particles in the room-air zone. Better visual but nothing outstandingly better	Clean, data-led layout with a native chart and icons. Less illustrative: simple silhouettes, no scene. More simplistic
Errors and defects	Major error on a value from the paper. 13x but the LLM wrote 1.3x	Even after multiple prompts with the LLM (namely on correcting the people shapes), defects are still present. Values are accurate	Values accurate. People-shape defects from 2 remain; the prompt restyled emphasis rather than fixing them.	Values match the paper. Added additional data on the dose.
Ease of editing	More difficult to request edits to the pixel images but once it was broken down in canvas AI, the text and illustrations were easy to edit	Easier to edit the individual shapes but it would not generate the same level of quality then a pixel-based image.	Same as 2: every shape and text box is editable.	Easiest: native text, icons and a chart whose data can be edited in PowerPoint.
Other comments	This option is much more token intensive (I assume). Hence, it is slightly more expensive	Shapes can be restyled with theme colours and fonts, which pixels cannot.	I was curious if the MS PowerPoint would improve the situation	Check how did Claude perform from scratch From one prompt: the only option that adds a plot